

Microfoundations II: Labor and Leisure

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Outline

- 1 Measuring the Labor Market
- 2 Income and Substitution Effects
- 3 Comparing the US and Europe
- 4 Labor Market Equilibrium



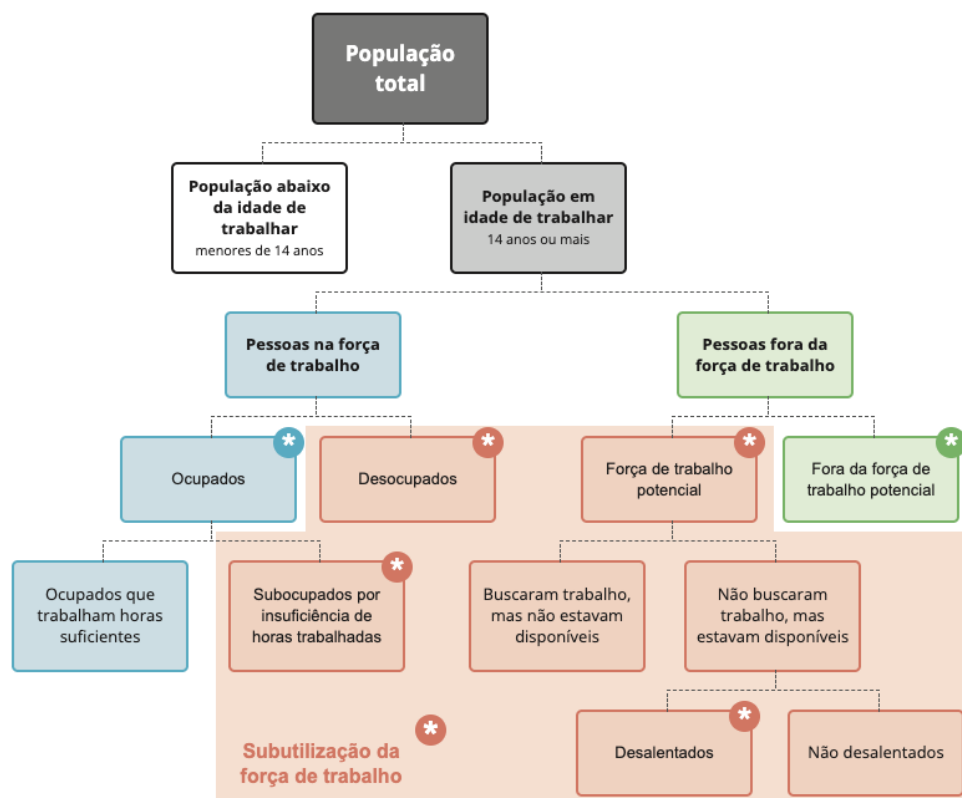
Labor Market Measures

- In labor market surveys individuals of the Working Age Population (15 to 64 years old OCDE; 14 up in Brazil) are classified as:
 - ▶ **Employed (E)**: worked in the past week
 - ▶ **Unemployed (U)**: not worked in the past week but actively searched for a job (i.e., was in the labor force)
 - ▶ **Out of the Labor Force**: didn't work and didn't search for a job in the past week
- Labor Force (LF) $\equiv E + U$

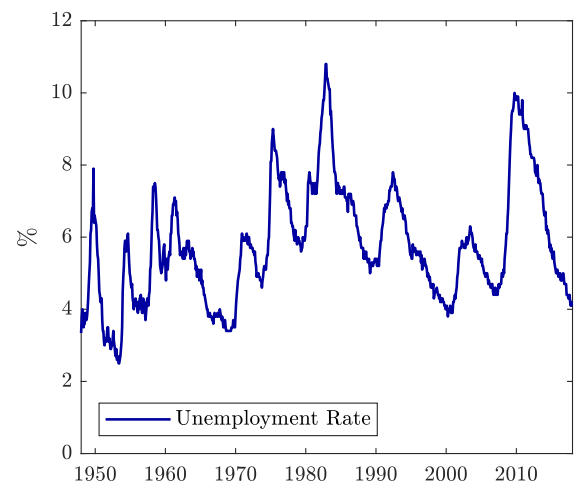
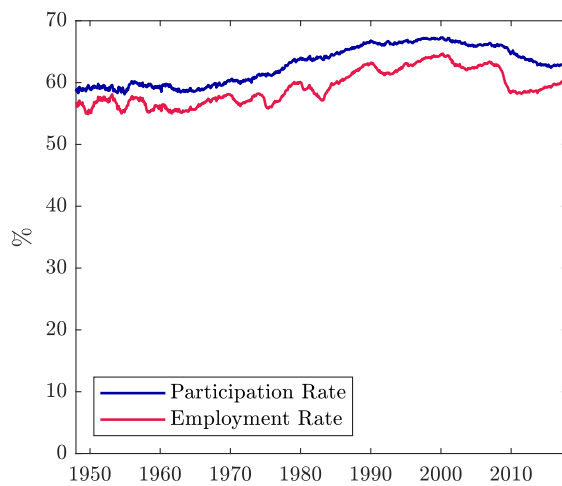
$$\text{Participation Rate} = \frac{LF}{\text{WA Population}}$$

$$\text{Employment Rate} = \frac{E}{WAPopulation}$$

$$\text{Unemployment Rate} = \frac{U}{LF}$$

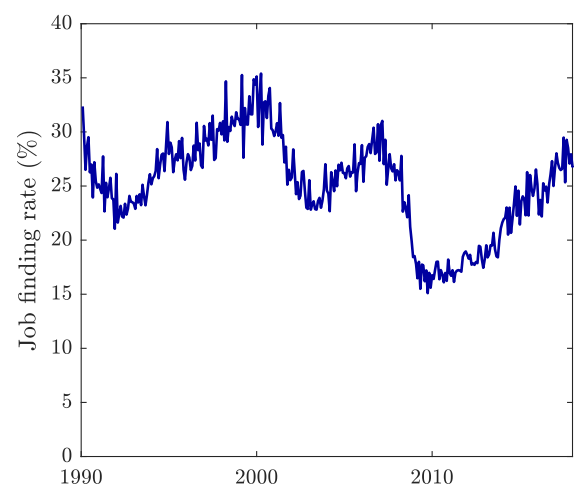
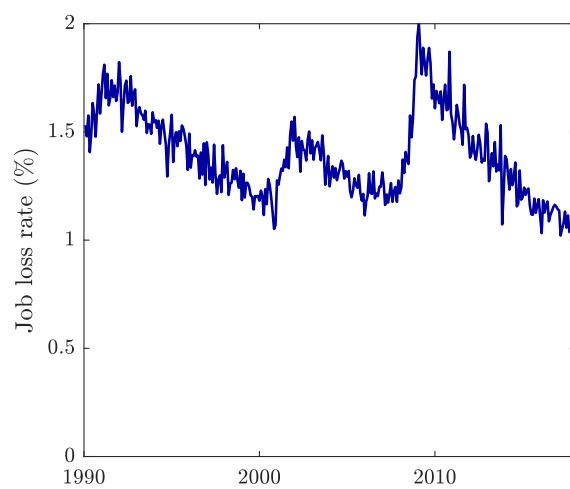


Labor Market Measures



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Labor Market Measures



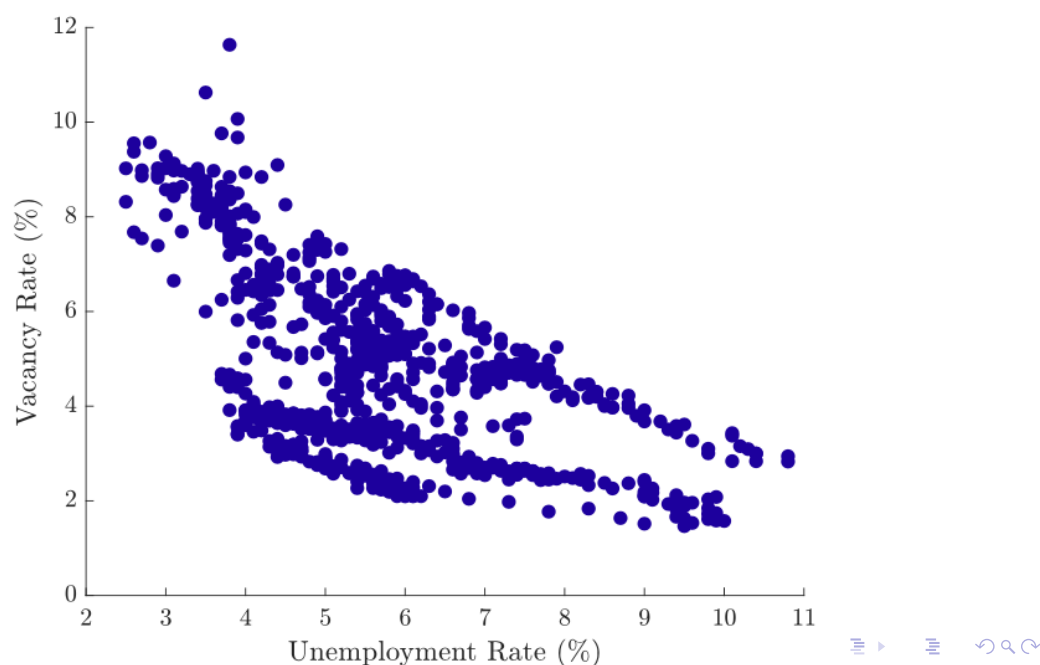
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Labor Market Measures

- **Vacancies (V)**: BLS asks firms how many job openings they have

$$\text{Vacancy Rate} = \frac{V}{LF}$$

- How does VR relate to unemployment? [**Beveridge Curve**]



Outline

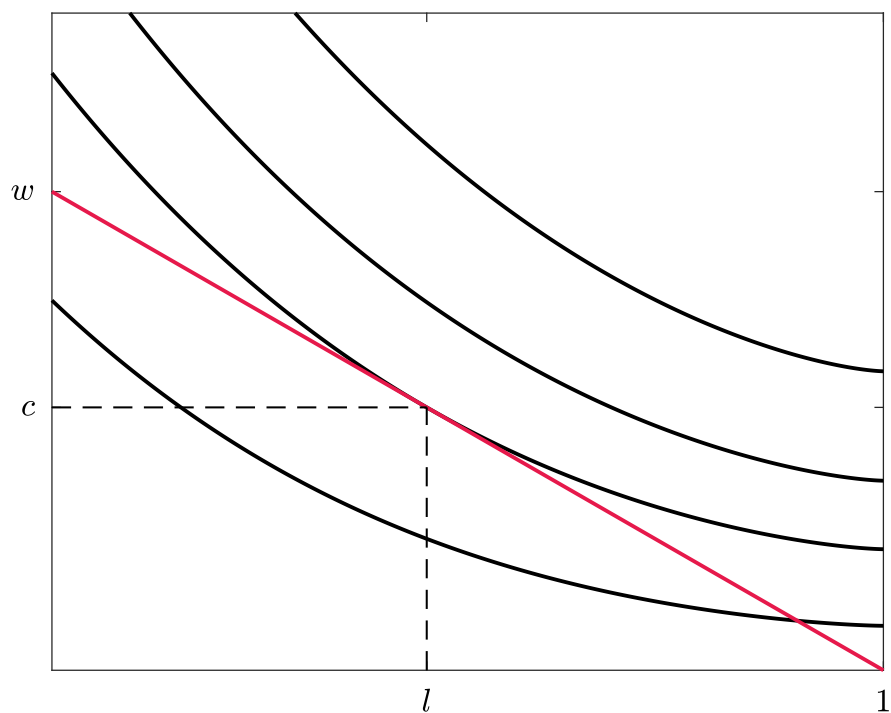
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Static Labor Supply Problem

- FOC:

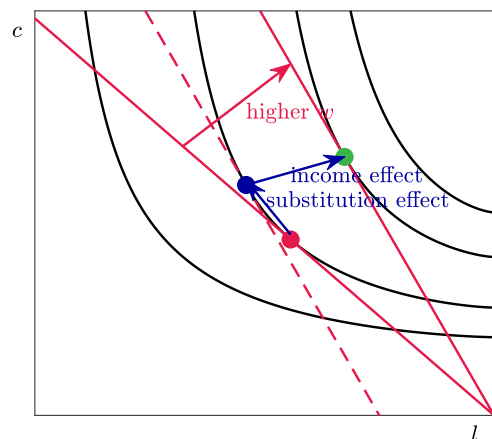
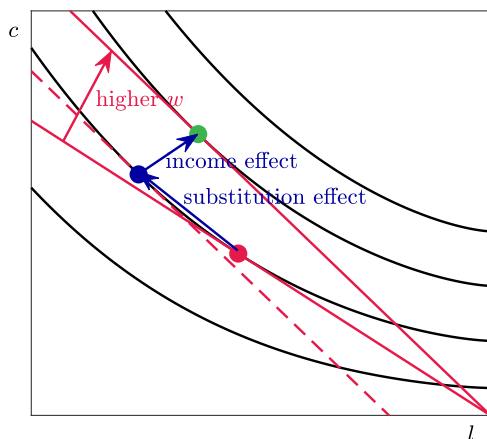
$$c \leq w(1 - l)$$

$$v'(l) = wu'(c)$$



Static Labor Supply Problem

- **Wage changes** bring Substitution and Wealth Effects
- Higher wages make:
 - ▶ Time more expensive, thus moving away from leisure and consuming more (relatively cheaper)
 - ▶ Workers richer: increase consumption, and leisure



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Static Labor Supply Problem

- Special case:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma} \quad v(l) = -\theta \frac{\epsilon}{1+\epsilon} (1-l)^{\frac{1+\epsilon}{\epsilon}}$$

- FOC:

$$\frac{\theta(1-l)^{\frac{1}{\epsilon}}}{c^{-\sigma}} = \frac{\theta(1-l)^{\frac{1}{\epsilon}}}{(w(1-l))^{-\sigma}} = w$$

- Solution:

$$\underbrace{1-l}_{\text{labor supply}} = \theta^{-\frac{1}{\frac{1}{\epsilon} + \sigma}} w^{\frac{1-\sigma}{\frac{1}{\epsilon} + \sigma}}$$

- Income vs. substitution depends on whether $\sigma > 1$.
 - ▶ Interpretation?

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Static Labor Supply Problem

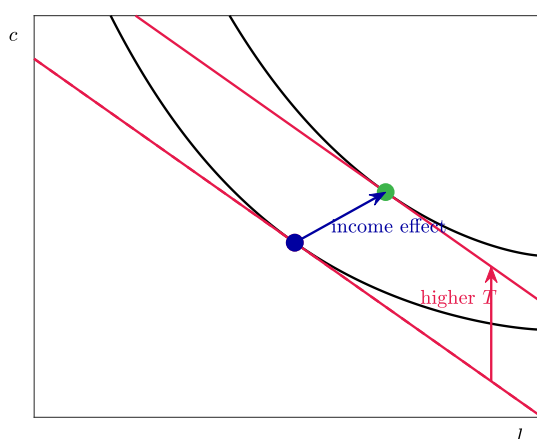
- **Adding Taxes and Transfers:** exogenously given quantities (worker's perspective), τ (income tax) and T (governmental transfer)
- New budget constraint:

$$c \leq w(1-l)(1-\tau) + T$$

- New FOC:

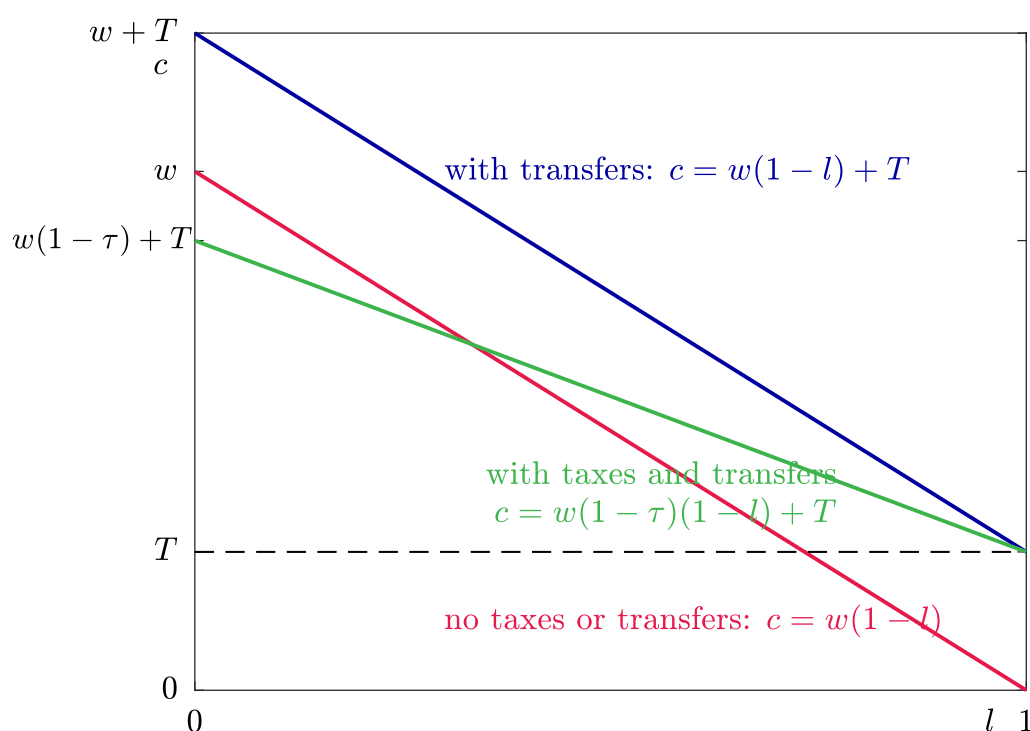
$$\frac{v'(l)}{u'(c)} = w(1-\tau)$$

- T has a purely income effect



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Static Labor Supply Problem



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Long-Term US Evidence

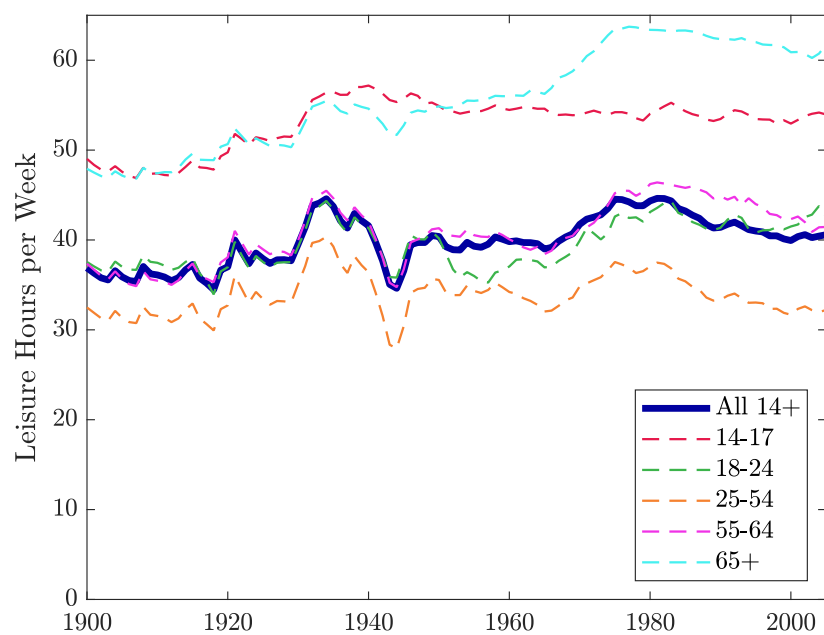


Figure: Ramey & Francis (2009)

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Long-Term US Evidence

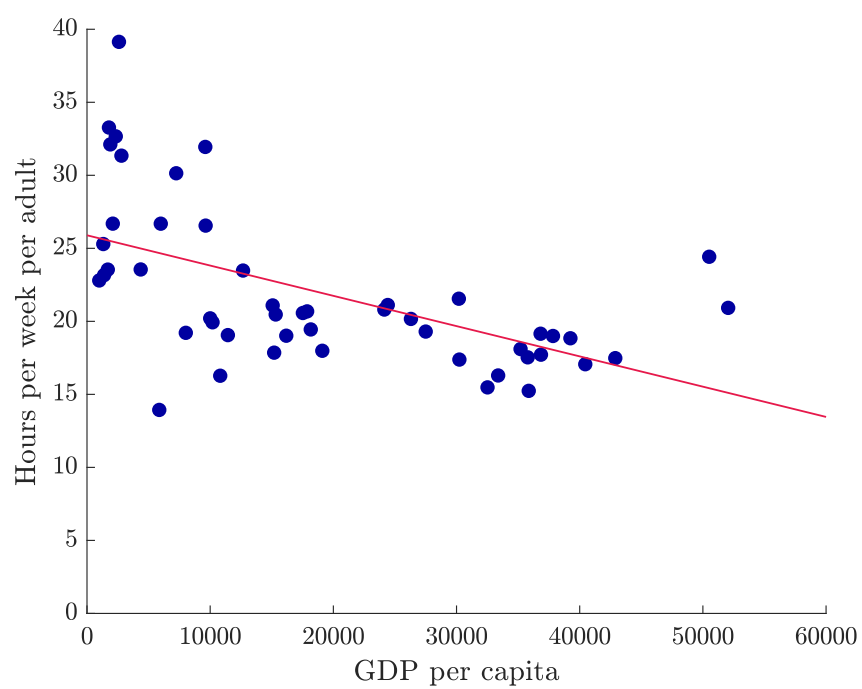
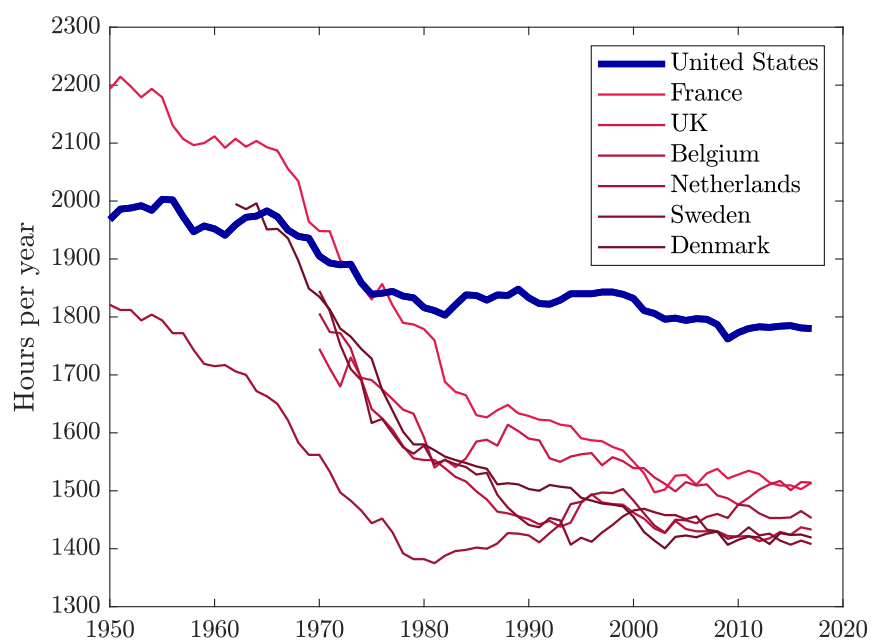


Figure: Bick et al. (2018)

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Is it Taxes?



Is it Unions?



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Competitive Labor Market

- We have seen that firms maximize profits and thus demand labor to make

$$w = F_L(K, L)$$

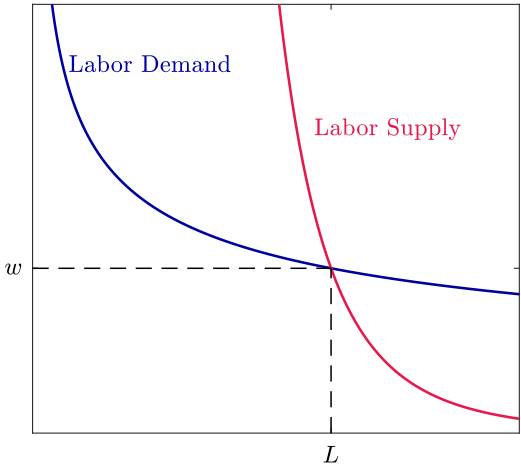
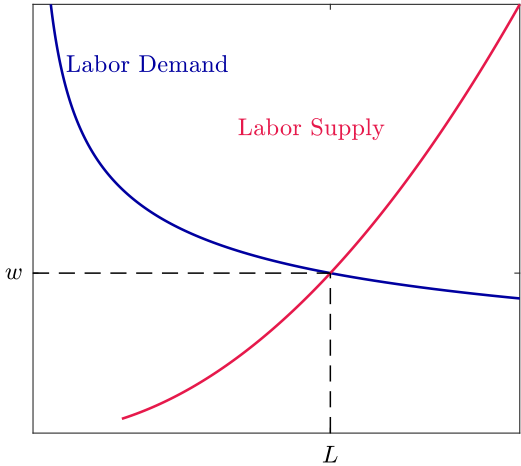
- Workers supply labor according to

$$\frac{v'(l)}{u'(w(1 - \tau)(1 - l) + T)} = w$$

- Labor demand is downward sloping
- Labor supply can be upward or downward sloping (substitution vs. income effect)

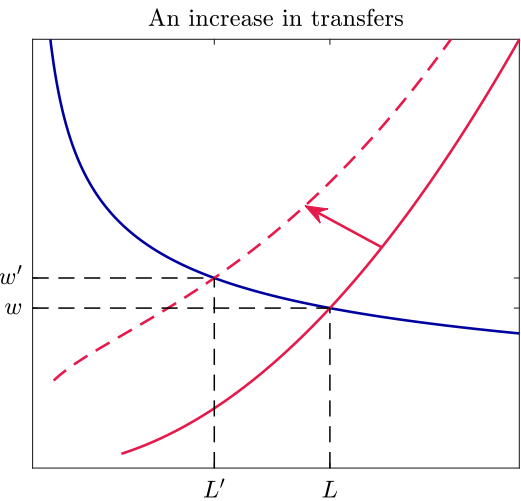
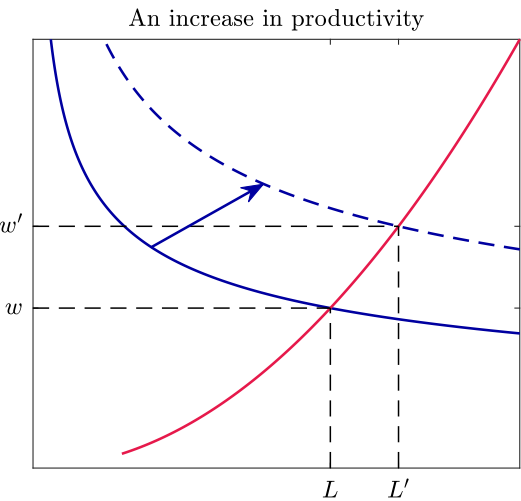
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Competitive Labor Market



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Competitive Labor Market



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Non-Competitive Labor Market: Search Model

- Competitive theory has no unemployment and no unfulfilled vacancies
- Thus cannot explain co-existence of unemployment and unfulfilled vacancies
- Search Model: Diamond (1982), Mortensen (1982), and Pissarides (1985)
- Searching for a job and finding a good fit for a vacancy take time and add frictions to the labor market
- To study this, let's abstract from labor supply: population is normalized to 1 and everybody work
- Production only happens after a worker is matched with a firm ("job")
- At the beginning of the period:
 - ▶ $1 - U$ workers with jobs and U unemployed searching for job
 - ▶ Firms advertise a vacancy at the cost χ units of output
 - ▶ Firms create a total of V vacancies
- Function $m(V, U)$ determines the number of U workers who find a job (**matching function**; increasing, concave in both U and V)

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Non-Competitive Labor Market: Search Model

- Define the vacancy filling rate:

$$q(V, U) = \frac{m(V, U)}{V}$$

which is increasing in U and decreasing in V (due to concavity of m)

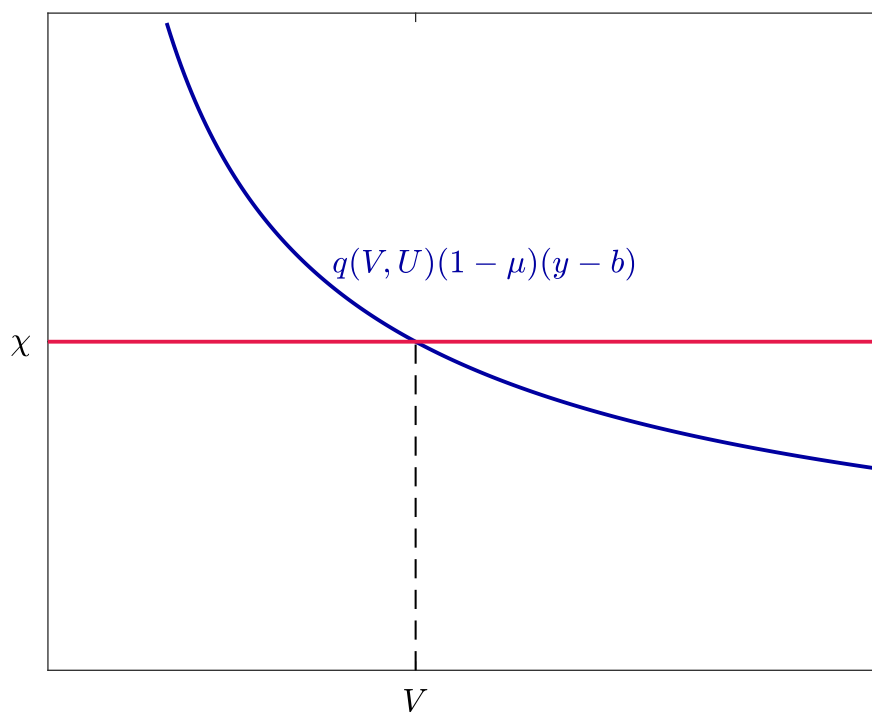
- Once there is a matching, workers and firms bargain the wage: **Nash bargaining**
 - 1 With no agreement: firm with unfulfilled vacancy and zero output and profits, and worker is unemployed and gets exogenous b (unemployment benefit or value of leisure)
 - 2 Reaching an agreement: they get y instead of b and thus share the surplus $y - b$ depending on the worker's bargaining power μ
 Wage will be: $w = b + \mu(y - b)$
 Profits will be $y - w = (1 - \mu)(y - b)$
- Firms' decision to post a vacancy: vacancy cost should equate the expected profit of hiring someone

$$\chi = q(V, U) \cdot (1 - \mu) \cdot (y - b) \quad (1)$$

solve (1) for V

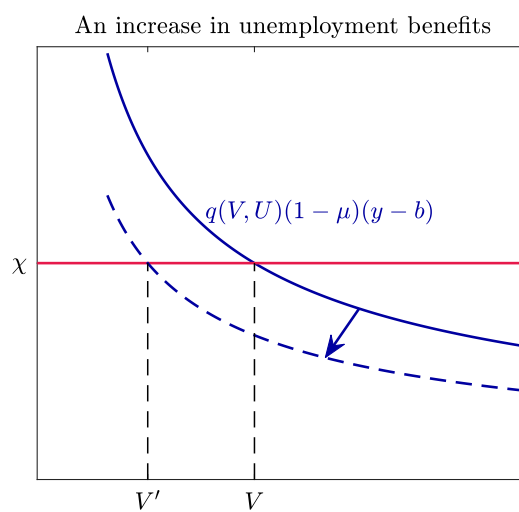
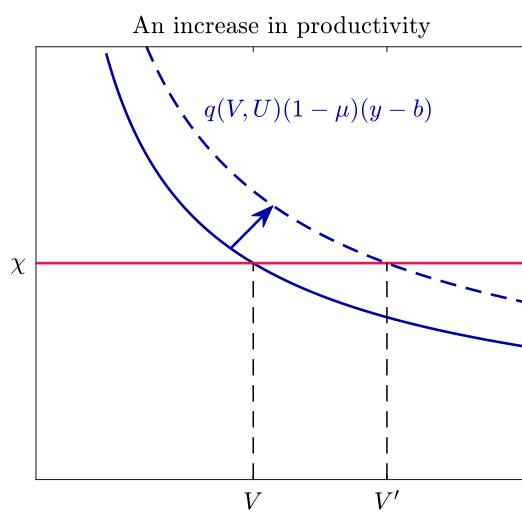
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Non-Competitive Labor Market: Search Model



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